

Credit Risk Modelling & Portfolio Stress Testing

PD, LGD, EAD, ECL, Monte Carlo VaR/ES and Stress Testing on 2.26 million LendingClub loans

Technical report: data, methodology, mathematics, statistical tests, results, and interview preparation

Executive summary

- **Data.** 2,260,701 accepted LendingClub loans (2007–2018Q4), snapshot March 2019. After cleaning, 2,257,919 loans; 1,762,677 have a complete 12-month outcome window (issued 2007–2017).
- **Design flaw found and fixed.** The original “terminal loans only” design is survivorship-biased: the 2017 terminal subset has a 12-month default rate of **15.1%** versus **5.8%** for the whole 2017 cohort (χ^2 $p \approx 0$). The target was re-specified as a censoring-free **12-month PD** (Basel/IFRS 9 horizon).
- **Out-of-time design.** Train 2007–2015 (884,691 loans), OOT test 2016 (434,407), OOT validation 2017 (443,579; \$6.585bn).
- **PD.** LightGBM champion: **AUC 0.725, Gini 0.450, KS 0.325** on 2017; +0.023 AUC over the WOE logistic scorecard (DeLong $p < 10^{-100}$).
- **Calibration.** Raw PDs under-predict (4.80% vs 5.80%); Platt + grade recalibration (chosen on 2016 by LR test, $p = 10^{-19}$) gives mean PD 5.90%, ECE 0.24 pp, Spiegelhalter $p = 0.076$; no red grade in the Jeffreys backtest; score PSI 0.009.
- **LGD / EAD.** LGD = 0.895 (pooled; no model beats the mean OOT), downturn LGD = 0.936; EAD fractional-logit model, OOT $R^2 = 0.29$.
- **ECL.** 12-month ECL **\$321.5M** (4.88%); realised \$326.1M (ratio 1.014). Lifetime ECL \$800.4M (12.15%).
- **Portfolio risk.** Loan-level one-factor Gaussian Monte Carlo (443,579 loans $\times$ 10,000 scenarios): EL \$320.5M, **99% VaR \$723.7M, 99% ES \$820.6M**, 99.9% VaR \$939.8M. Combined adverse stress: 99% VaR \$1,172.3M, ES \$1,343.1M. Reverse stress: PD odds $\times 3.74$ make EL equal to base 99.9% VaR.

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1 Data

1.1 Source and cleaning

- Source: Kaggle [wordsofthewise/lending-club](#), file `accepted_2007_to_2018Q4.csv.gz`: **2,260,701 rows, 151 columns**; issue dates June 2007 – December 2018; the latest payment date in the file is **March 2019** (data snapshot).
- Removed 33 rows with missing status/issue date and 2,749 “Does not meet the credit policy” loans (a different underwriting regime, 2007–2010): **2,257,919 loans** remain.
- Loan status at snapshot: Fully Paid 1,076,751; Current 878,317; Charged Off 268,559; Late (31–120) 21,467; In Grace 8,436; Late (16–30) 4,349; Default 40.
- Columns used: 67 application-time fields (model inputs) and 9 outcome fields (targets, LGD, EAD only).

Leakage control. Only information known at origination is a model input. Post-origination fields (payments, recoveries, `last_fico`, `out_prncp`, hardship and settlement flags, `last_pymnt_d`, ...) are used *only* to construct targets, LGD and EAD.

Table 1: Cohort summary. “Terminal” = Fully Paid / Charged Off / Default at the snapshot.

Issue year	Loans	Funded (\$bn)	Terminal loans	Lifetime DR (terminal)	12m DR (all loans)
2007.0	251	0.002	251	17.93%	5.18%
2008.0	1,562	0.013	1,562	15.81%	5.51%
2009.0	4,716	0.046	4,716	12.60%	5.53%
2010.0	11,536	0.117	11,536	12.89%	4.31%
2011.0	21,721	0.257	21,721	15.18%	4.87%
2012.0	53,367	0.718	53,367	16.20%	5.26%
2013.0	134,814	1.983	134,804	15.60%	4.40%
2014.0	235,629	3.504	223,103	18.45%	4.75%
2015.0	421,095	6.418	375,546	20.19%	5.51%
2016.0	434,407	6.401	293,105	23.29%	6.28%
2017.0	443,579	6.585	169,321	23.13%	5.80%
2018.0	495,242	7.936	56,318	15.76%	–

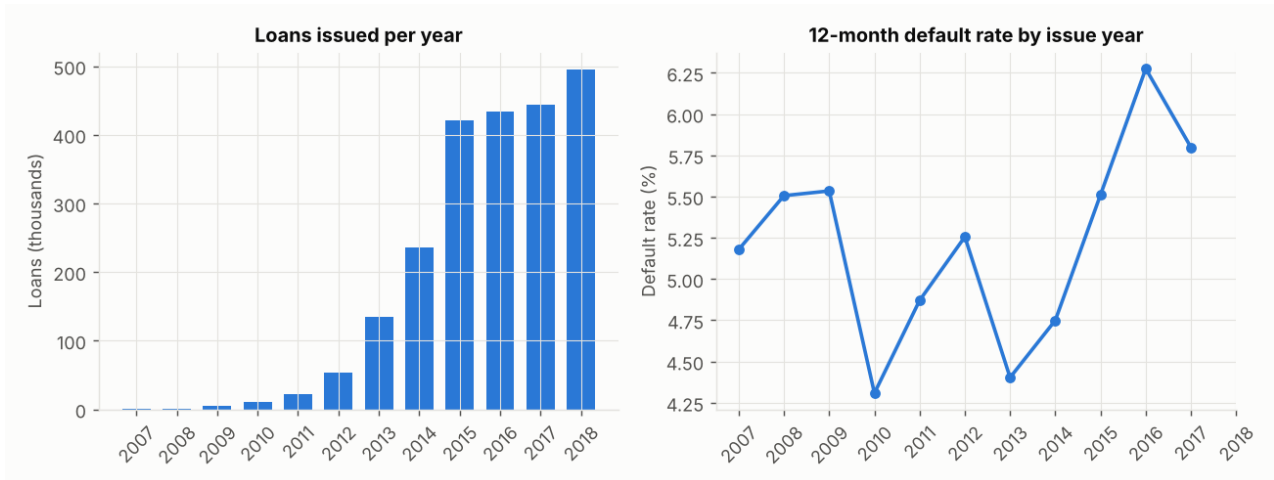


Figure 1: Loan volume and 12-month default rate by issue year (2018 has no complete 12-month window).

1.2 Target definition: 12-month default

Let $\mathcal{B} = \{\text{Charged Off, Default, Late (31–120 days)}\}$ be the set of default statuses. For loan i with issue month I_i , last-payment month LP_i , recoveries R_i , principal and interest received P_i , Int_i and instalment A_i , the month-on-book of the first missed instalment is

$$\tau_i = \text{clip} \left(\begin{cases} m(LP_i) - m(I_i) + 1, & R_i = 0 \text{ and } LP_i \text{ observed,} \\ \left\lceil \frac{P_i + Int_i}{A_i} \right\rceil + 1, & \text{otherwise,} \end{cases}, 1, \text{term}_i \right) \quad (1)$$

The payment-count rule is used when recovery cash may have moved the last-payment date. On the 104,436 charge-offs without recoveries it agrees with the date rule on the 12-month flag in **98.0%** of cases (mean bias -0.11 months). The target is

$$D_i = \mathbf{1}\{\text{status}_i \in \mathcal{B}\} \cdot \mathbf{1}\{\tau_i \leq 12\}, \quad \text{defined for every loan with } I_i \leq \text{Dec 2017}. \quad (2)$$

Loans issued in December 2017 have their 12-month window close in December 2018; a miss at month 12 is ≥ 90 days past due by the March 2019 snapshot and is captured as Late (31–120) or Charged Off. Prepayment or survival beyond 12 months is $D_i = 0$, so no loan is excluded because it is still open: **the target is free of survivorship bias**.

1.3 Why the original design was replaced

The original project used only *terminal* loans with a lifetime target. For recent cohorts, only loans that had already ended are included, and those are disproportionately early charge-offs and early prepayments.

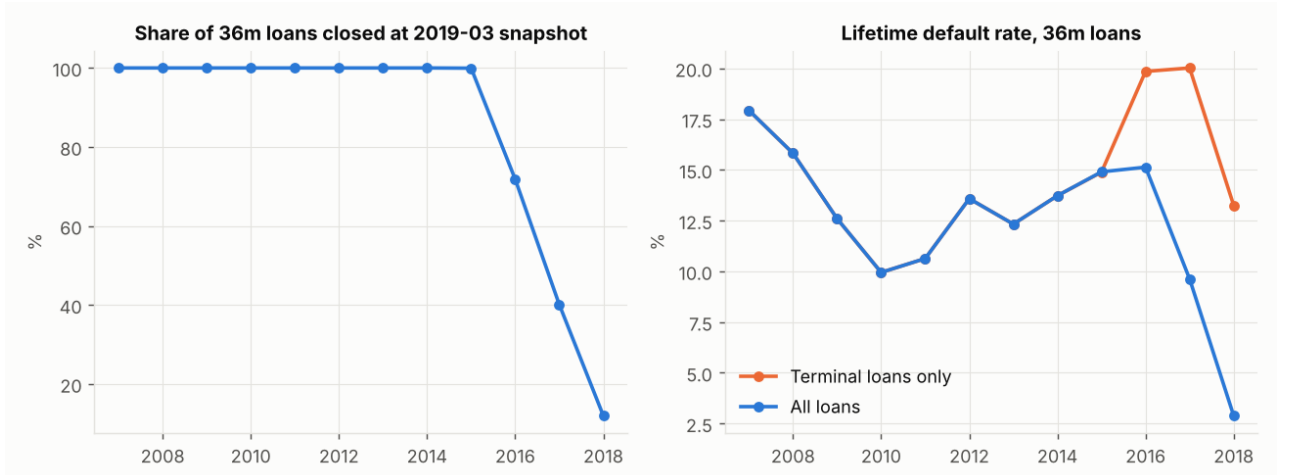


Figure 2: 36-month loans: once a cohort is fully matured (2007–2015) the terminal and full default rates coincide; for 2016–2017 the terminal subset strongly over-states default.

- 2017: 169,321 terminal loans = **38.2%** of loans (36.8% of dollars).
- 12-month default rate: terminal subset **15.08%** vs whole cohort **5.80%**; $2 \times 2 \chi^2$ test terminal vs non-terminal: $p \approx 0$.
- 76.9% of the 2017 terminal loans are full early repayments; the remainder are early defaults.

Replicating the original design gives AUC 0.693 (LR) / 0.721 (LightGBM) and ECL \$454.2M on \$2.42bn, close to the original claim (AUC 0.700, ECL \$497.3M) – but these figures describe a censored, selected sample, not the 2017 portfolio.

2 Sample design (out-of-time)

Table 2: Development, test and validation samples (12-month target).

Sample	Issue years	Loans	12m default rate	Role
Development (train)	2007–2015	884,691	5.09%	model fitting
tuning split A	2007–2014	463,596	–	fit during hyper-parameter search
tuning split B	2015	421,095	5.51%	early stopping / selection (time-based)
OOT test	2016	434,407	6.28%	champion choice, recalibration, stacking
OOT validation	2017	443,579	5.80%	final evaluation (scored once)

Protocol. Every decision (variables, hyper-parameters, champion, recalibration method, stacking weights) uses data up to 2016 only. The 2017 cohort is never used for a choice.

Training window test. Bureau fields (e.g. `bc_util`, `tot_cur_bal`) are 100% missing before 2012. To check whether old vintages harm the model, LightGBM was trained on four windows ending 2014 and scored on 2015; paired DeLong tests show no significant difference, so the full window is kept.

Table 3: Training-window experiment (fixed LightGBM parameters).

Train window	n train	AUC 2015	95% CI	DeLong p vs 2007–2014
2007-2014	463,596	0.7324	[0.7293, 0.7356]	ref.
2010-2014	457,067	0.7327	[0.7295, 0.7358]	0.5202
2012-2014	423,810	0.7331	[0.7299, 0.7362]	0.0966
2013-2014	370,443	0.7328	[0.7296, 0.7359]	0.4124

3 Features and feature selection

3.1 Candidate features

68 candidates: 64 numeric and 4 categorical (`home_ownership`, `verification_status`, `purpose`, `addr_state`). Engineered features:

$$\begin{aligned}
 \text{credit_hist_m} &= m(I_i) - m(\text{earliest credit line}), & \text{fico} &= \frac{1}{2}(\text{fico_low} + \text{fico_high}), \\
 \text{loan_to_inc} &= \frac{\text{loan amount}}{\text{annual income}}, & \text{inst_to_inc} &= \frac{12 \cdot \text{instalment}}{\text{annual income}}, \\
 \text{revol_to_inc} &= \frac{\text{revolving balance}}{\text{annual income}}, & \text{log_inc} &= \ln(1 + \text{annual income}),
 \end{aligned}$$

plus `sub_grade_n` (A1= 1, ..., G5= 35), employment length in years with a missing flag, joint-application and whole-loan flags; DTI outside $[0, 100]$ is set missing.

3.2 Weight of evidence and information value

For a binned variable with bins $b = 1, \dots, K$, B_b defaults and G_b non-defaults (with +0.5 smoothing),

$$\text{WOE}_b = \ln \frac{B_b/B}{G_b/G}, \quad \text{IV} = \sum_{b=1}^K \left(\frac{B_b}{B} - \frac{G_b}{G} \right) \text{WOE}_b. \quad (3)$$

Higher WOE means riskier, so every scorecard coefficient is expected to be positive.

Monotone binning. Numeric variables start from 20 quantile bins; adjacent bins are merged until (i) the default rate is monotone in the direction of the Spearman correlation between x and D and (ii)

every bin holds $\geq 5\%$ of the sample. Missing values form their own bin (WOE set to 0 if $< 0.5\%$). Categorical levels below 0.5% are pooled into “OTHER”.

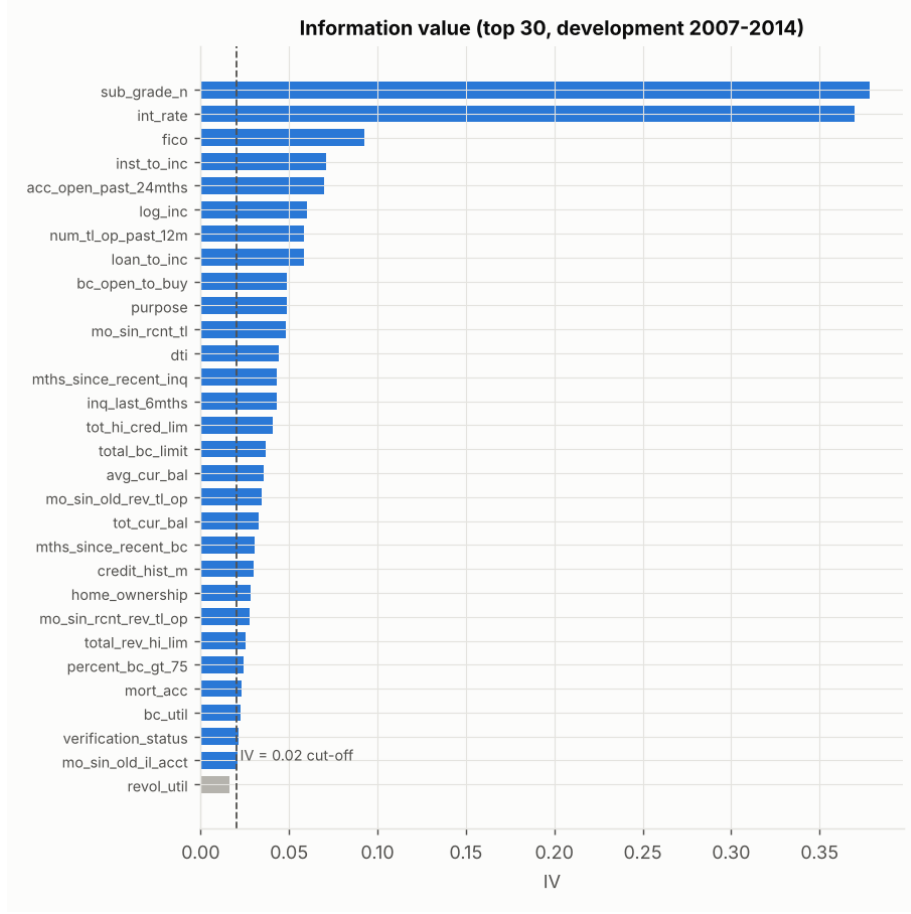


Figure 3: Information value of the top 30 candidates (development 2007–2014). 29 variables pass $IV \geq 0.02$.

3.3 Selection funnel for the logistic scorecard

1. **IV filter:** $IV \geq 0.02 \Rightarrow 29$ variables.
2. **Correlation filter:** in IV order, drop a variable if $|r| > 0.7$ with a stronger kept one $\Rightarrow 21$ variables (e.g. `int_rate` is dropped because it is collinear with `sub_grade_n`).
3. **Forward stepwise selection** on 2007–2014, holdout 2015. At each step every remaining variable is tried; the best is accepted only if *all* gates pass:
 - AUC gain on the 2015 holdout $\geq g_{\min}$ (5×10^{-4} for v1, 10^{-4} for v2);
 - Wald test $p < 0.001$;
 - all coefficients positive (WOE sign consistency);
 - all $VIF < 5$, where $VIF_j = 1/(1 - R_j^2)$ and R_j^2 is from regressing WOE_j on the other WOE's.

Table 4: Forward stepwise path (scorecard v2). v1 stops after step 5.

Step	Variable added	Holdout AUC	AUC gain	Wald p
1	sub_grade_n	0.7042	+0.2042	$< 10^{-299}$
2	acc_open_past_24mths	0.7130	+0.0088	$< 10^{-140}$
3	home_ownership	0.7174	+0.0044	$< 10^{-100}$
4	dti	0.7184	+0.0010	$< 10^{-50}$
5	mths_since_recent_inq	0.7196	+0.0011	$< 10^{-28}$
6	mort_acc	0.7200	+0.0004	$< 10^{-22}$
7	mo_sin_old_il_acct	0.7202	+0.0002	$< 10^{-45}$
8	verification_status	0.7205	+0.0003	0.0006
9	mo_sin_old_rev_tl_op	0.7207	+0.0001	$< 10^{-24}$
10	inq_last_6mths	0.7208	+0.0001	$< 10^{-24}$
11	mo_sin_rcnt_tl	0.7209	+0.0001	$< 10^{-8}$
12	total_rev_hi_lim	0.7210	+0.0001	$< 10^{-9}$

4 Models

4.1 Logistic regression scorecard

$$\text{logit } P(D_i = 1 \mid \mathbf{x}_i) = \beta_0 + \sum_{j \in S} \beta_j \text{WOE}_j(x_{ij}), \quad \hat{\beta} = \arg \max_{\beta} \sum_i [D_i \ln p_i + (1 - D_i) \ln(1 - p_i)]. \quad (4)$$

Inference: Wald $z_j = \hat{\beta}_j / \widehat{SE}(\hat{\beta}_j)$; nested models are compared with the likelihood-ratio statistic $\Lambda = 2(\ell_1 - \ell_0) \sim \chi^2_{\Delta k}$; information criteria $\text{AIC} = 2k - 2\ell$, $\text{BIC} = k \ln n - 2\ell$.

Table 5: Scorecard v1 (5 WOE variables), fitted on 2007–2015. LR test vs null $\chi^2 = 22,585$, pseudo- $R^2 = 0.063$.

Term	$\hat{\beta}$	SE	z	p	VIF
const	-2.9268	0.0052	-563.5	$< 10^{-299}$	–
sub_grade_n	0.8930	0.0082	109.1	$< 10^{-299}$	1.10
acc_open_past_24mths	0.5780	0.0161	35.9	$< 10^{-282}$	1.12
home_ownership	0.9144	0.0272	33.7	$< 10^{-247}$	1.02
dti	0.4404	0.0200	22.0	$< 10^{-106}$	1.05
mths_since_recent_inq	0.3340	0.0217	15.4	$< 10^{-52}$	1.11

Table 6: Scorecard v2 (12 WOE variables), fitted on 2007–2015.

Term	$\hat{\beta}$	SE	z	p	VIF
const	-2.9273	0.0052	-562.3	$< 10^{-299}$	–
sub_grade_n	0.8372	0.0086	97.2	$< 10^{-299}$	1.29
acc_open_past_24mths	0.5031	0.0183	27.5	$< 10^{-166}$	1.50
home_ownership	0.6886	0.0308	22.3	$< 10^{-109}$	1.31
dti	0.4948	0.0205	24.1	$< 10^{-127}$	1.10
mths_since_recent_inq	0.2078	0.0242	8.6	$< 10^{-17}$	1.33
mort_acc	0.2031	0.0364	5.6	$< 10^{-7}$	1.44
mo_sin_old_il_acct	0.3825	0.0363	10.5	$< 10^{-25}$	1.15
verification_status	0.2559	0.0247	10.4	$< 10^{-24}$	1.09
mo_sin_old_rev_tl_op	0.3179	0.0272	11.7	$< 10^{-30}$	1.33
inq_last_6mths	0.2708	0.0250	10.8	$< 10^{-26}$	1.23
mo_sin_rcnt_tl	0.1520	0.0245	6.2	$< 10^{-9}$	1.49
total_rev_hi_lim	0.1891	0.0335	5.6	$< 10^{-7}$	1.28

v2 vs v1. Nested LR test $\chi^2 = 792.8$, $df = 7$, $p = 6.7 \times 10^{-167}$; AIC $333,498 \rightarrow 332,719$; BIC $333,568 \rightarrow 332,871$; DeLong on 2016: $\Delta\text{AUC} = +0.0011$, $z = 3.40$, $p = 0.0007$. v2 is statistically better but the economic gain is small.

Scorecard points. With PDO = 20, 600 points at good:bad odds 50:1,

$$\text{Score} = \text{Offset} + \text{Factor} \cdot \ln(\text{odds}_{\text{good}}), \quad \text{Factor} = \frac{\text{PDO}}{\ln 2}, \quad \text{Offset} = 600 - \text{Factor} \ln 50, \quad (5)$$

and the points for attribute b of variable j are $-(\beta_j \text{WOE}_{jb} + \beta_0/k) \text{Factor} + \text{Offset}/k$ (table 02_scorecard_points.csv).

4.2 Machine-learning challengers

Gradient boosting builds an additive model $F_M(\mathbf{x}) = \sum_{m=1}^M \eta f_m(\mathbf{x})$ of trees f_m minimising the regularised log-loss

$$\mathcal{L} = \sum_i \ell(D_i, F(\mathbf{x}_i)) + \sum_m \Omega(f_m), \quad \Omega(f) = \gamma T + \frac{1}{2} \lambda \|\mathbf{w}\|^2, \quad (6)$$

with ℓ the binary cross-entropy, T leaves and leaf weights $\mathbf{w}$. Each tree fits the gradient $g_i = p_i - D_i$ and Hessian $h_i = p_i(1 - p_i)$; the optimal leaf weight is $w_j^* = -\sum_{i \in j} g_i / (\sum_{i \in j} h_i + \lambda)$. LightGBM grows trees leaf-wise on histograms with native categorical splits; XGBoost grows depth-wise. **Random forest:** average of 500 trees on 30% bootstrap samples with random feature subsets.

Tuning: random search (LightGBM 20, XGBoost 12, RF 6 configurations), fitted on 2007–2014 with early stopping (100 rounds) on 2015; the selected number of rounds is scaled by $n_{2007-15}/n_{2007-14} = 1.91$ for the final refit.

Table 7: Selected hyper-parameters. Best 2015 holdout AUC: LightGBM 0.7355, XGBoost 0.7344, RF 0.7246.

Model	Selected hyper-parameters (refit on 2007–2015)
LightGBM	num_leaves=63, min_child_samples=200, learning_rate=0.03, feature_fraction=0.5, bagging_fraction=0.7, lambda_l2=50.0, min_split_gain=0.0, n_iter=1200
XGBoost	max_depth=8, min_child_weight=50.0, learning_rate=0.03, subsample=0.9, colsample_bytree=0.5, reg_lambda=1.0, n_iter=849
Random Forest	min_samples_leaf=200, max_features=0.2, n_estimators=500, max_samples=0.3

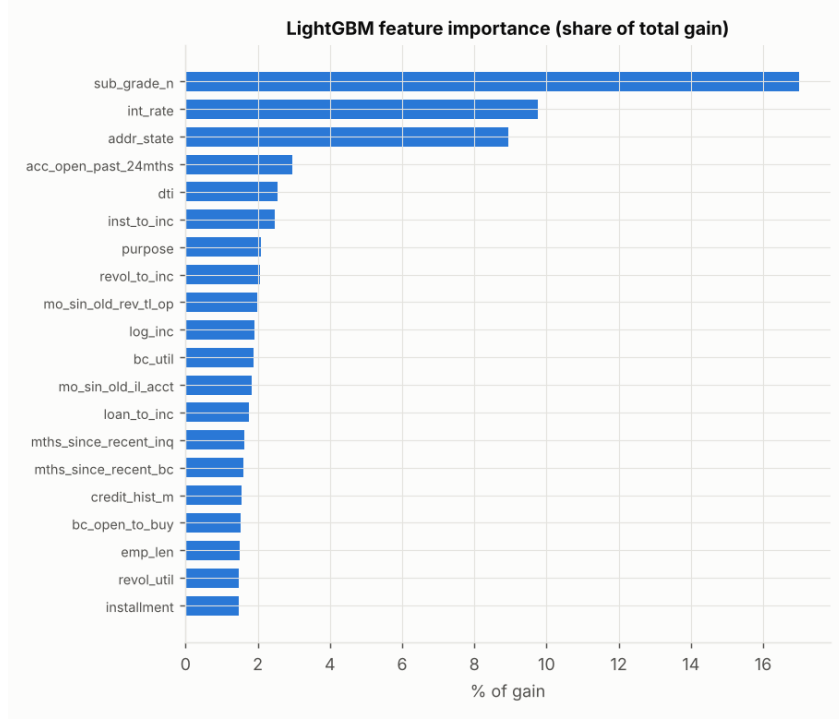


Figure 4: LightGBM gain importance. Sub-grade, interest rate and state dominate; LendingClub’s own grade already prices FICO and much of the bureau data.

Stacking: $\text{logit } \hat{p} = \gamma_0 + \gamma_1 \text{logit } p_{LR} + \gamma_2 \text{logit } p_{LGBM} + \gamma_3 \text{logit } p_{XGB}$ fitted on 2016; $\hat{\gamma} = (0.194, 0.136, 0.549, 0.295)$, all $p < 10^{-13}$.

5 Discrimination

5.1 Definitions

For scores s of defaulters (m loans) and non-defaulters (n loans),

$$\text{AUC} = P(s_D > s_{ND}) + \frac{1}{2}P(s_D = s_{ND}) = \frac{1}{mn} \sum_{i \in D} \sum_{j \in ND} \psi(s_i, s_j), \quad \psi = \mathbf{1}\{s_i > s_j\} + \frac{1}{2}\mathbf{1}\{s_i = s_j\}, \quad (7)$$

$$\text{Gini} = 2 \text{AUC} - 1, \quad \text{KS} = \max_t |F_{ND}(t) - F_D(t)|, \quad (8)$$

$$\text{Brier} = \frac{1}{N} \sum_i (\hat{p}_i - D_i)^2, \quad \text{Log-loss} = -\frac{1}{N} \sum_i [D_i \ln \hat{p}_i + (1 - D_i) \ln(1 - \hat{p}_i)]. \quad (9)$$

DeLong test. With structural components $V_{10}(s_i) = \frac{1}{n} \sum_j \psi(s_i, s_j)$ and $V_{01}(s_j) = \frac{1}{m} \sum_i \psi(s_i, s_j)$, the covariance of AUC estimates for models r, s is

$$\widehat{\text{Cov}}(\hat{A}_r, \hat{A}_s) = \frac{S_{10}^{rs}}{m} + \frac{S_{01}^{rs}}{n}, \quad z = \frac{\hat{A}_1 - \hat{A}_2}{\sqrt{\text{Var}(\hat{A}_1) + \text{Var}(\hat{A}_2) - 2\text{Cov}(\hat{A}_1, \hat{A}_2)}} \sim N(0, 1), \quad (10)$$

computed with the $O(N \log N)$ algorithm of Sun & Xu (2014). With 15 pairwise tests, p -values are Holm-adjusted: $p_{(k)}^{adj} = \max_{j \leq k} \min\{1, (M - j + 1)p_{(j)}\}$.

5.2 Results

Table 8: Out-of-time discrimination of the raw models (2016 test, 2017 validation).

Sample	Model	AUC	95% DeLong CI	Gini	KS	Brier	Log-loss
test 2016	LogReg_WOE	0.7108	[0.7077, 0.7138]	0.4216	0.3058	0.0569	0.2190
test 2016	LightGBM	0.7326	[0.7297, 0.7356]	0.4652	0.3396	0.0561	0.2146
test 2016	XGBoost	0.7313	[0.7284, 0.7343]	0.4626	0.3373	0.0562	0.2151
test 2016	RandomForest	0.7209	[0.7179, 0.7239]	0.4418	0.3212	0.0566	0.2170
valid 2017	LogReg_WOE	0.7021	[0.6989, 0.7053]	0.4042	0.2915	0.0531	0.2080
valid 2017	LightGBM	0.7252	[0.7221, 0.7282]	0.4503	0.3253	0.0524	0.2041
valid 2017	XGBoost	0.7238	[0.7208, 0.7269]	0.4477	0.3229	0.0525	0.2045
valid 2017	RandomForest	0.7144	[0.7112, 0.7175]	0.4287	0.3091	0.0527	0.2056

Table 9: 2017 validation including scorecard v2 and stacking.

Model (2017)	AUC	95% CI	Gini	KS
LogReg_WOE	0.7021	[0.6989, 0.7053]	0.4042	0.2915
LightGBM	0.7252	[0.7221, 0.7282]	0.4503	0.3253
XGBoost	0.7238	[0.7208, 0.7269]	0.4477	0.3229
RandomForest	0.7144	[0.7112, 0.7175]	0.4287	0.3091
LogReg_WOE_v2	0.7035	[0.7004, 0.7067]	0.4071	0.2956
Stacked	0.7260	[0.7230, 0.7291]	0.4521	0.3266

Table 10: Paired DeLong tests on 2017 with Holm correction ($\Delta\text{AUC} = A - B$).

Comparison (A vs B)	ΔAUC	z	p	p (Holm)
LogReg_WOE vs LightGBM	-0.0231	-24.03	$< 10^{-126}$	$< 10^{-125}$
LogReg_WOE vs XGBoost	-0.0217	-22.44	$< 10^{-110}$	$< 10^{-109}$
LogReg_WOE vs RandomForest	-0.0123	-20.47	$< 10^{-92}$	$< 10^{-91}$
LogReg_WOE vs LogReg_WOE_v2	-0.0014	-4.14	$< 10^{-4}$	$< 10^{-4}$
LogReg_WOE vs Stacked	-0.0239	-27.90	$< 10^{-170}$	$< 10^{-169}$
LightGBM vs XGBoost	+0.0013	3.65	0.0003	0.0003
LightGBM vs RandomForest	+0.0108	14.69	$< 10^{-48}$	$< 10^{-47}$
LightGBM vs LogReg_WOE_v2	+0.0216	23.34	$< 10^{-119}$	$< 10^{-118}$
LightGBM vs Stacked	-0.0009	-5.23	$< 10^{-6}$	$< 10^{-6}$
XGBoost vs RandomForest	+0.0095	12.71	$< 10^{-36}$	$< 10^{-35}$
XGBoost vs LogReg_WOE_v2	+0.0203	21.77	$< 10^{-104}$	$< 10^{-103}$
XGBoost vs Stacked	-0.0022	-8.54	$< 10^{-16}$	$< 10^{-16}$
RandomForest vs LogReg_WOE_v2	+0.0108	19.52	$< 10^{-84}$	$< 10^{-83}$
RandomForest vs Stacked	-0.0117	-18.16	$< 10^{-72}$	$< 10^{-72}$
LogReg_WOE_v2 vs Stacked	-0.0225	-27.62	$< 10^{-167}$	$< 10^{-165}$

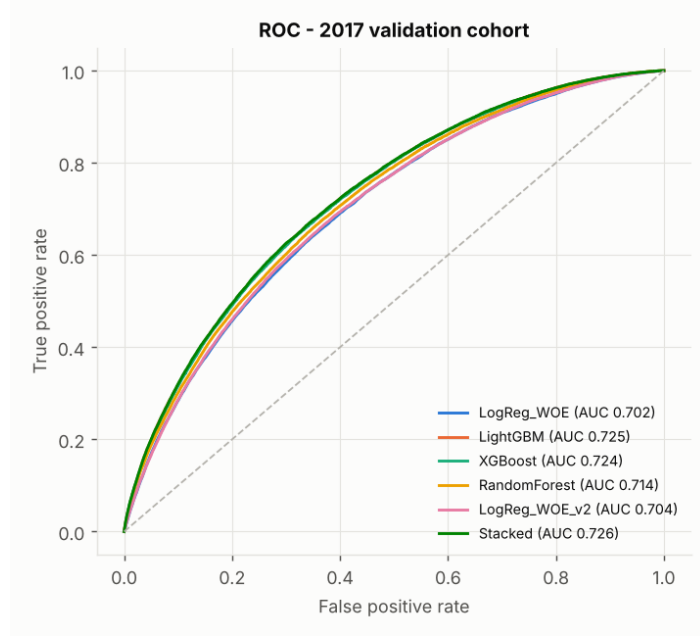


Figure 5: ROC curves on the 2017 validation cohort.

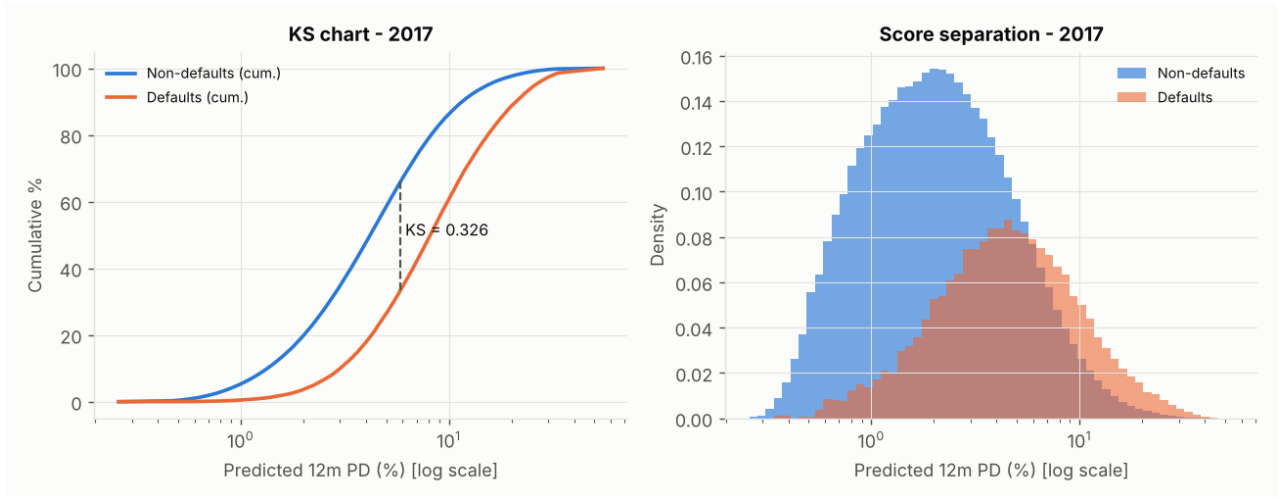


Figure 6: KS chart and score separation of the final PD, 2017.

Conclusion. The champion was chosen on 2016 AUC (LightGBM 0.7326). On 2017 it beats the scorecard by +0.023 AUC (LR v1) and +0.022 (LR v2), XGBoost by +0.0013 ($p_{Holm} = 0.0003$) and RF by +0.011, all significant. Stacking adds +0.0009 AUC: significant but immaterial, so the single LightGBM is retained for simplicity.

6 Calibration and recalibration

6.1 Calibration statistics

- **Calibration intercept/slope** (logistic recalibration): $\text{logit } P(D = 1) = a + b \text{ logit } \hat{p}$. Calibration-in-the-large (CITL) is a with b fixed to 1 (offset). Ideal: $a = 0$, $b = 1$; Wald tests $H_0 : a = 0$ and $H_0 : b = 1$.

- **Hosmer–Lemeshow** over $G = 10$ PD deciles:
$$\text{HL} = \sum_{g=1}^G \frac{(O_g - E_g)^2}{N_g \bar{p}_g (1 - \bar{p}_g)} \sim \chi_{G-2}^2.$$

- **Spiegelhalter** $Z = \frac{\sum_i (D_i - \hat{p}_i)(1 - 2\hat{p}_i)}{\sqrt{\sum_i (1 - 2\hat{p}_i)^2 \hat{p}_i (1 - \hat{p}_i)}} \sim N(0, 1)$.
- **Expected calibration error** over 20 equal-frequency bins: $ECE = \sum_b \frac{N_b}{N} |\bar{D}_b - \bar{p}_b|$.

With $N = 443,579$, HL has very high power and rejects deviations of a few basis points, so it is always reported alongside effect sizes (ECE, CITL).

6.2 Recalibration methods (fitted on 2016, applied to 2017)

$$\text{Intercept shift: } \hat{p}' = \Lambda(\text{logit } \hat{p} + a), \quad a \text{ solves } \frac{1}{N} \sum_i \Lambda(\text{logit } \hat{p}_i + a) = \bar{D}_{2016}, \quad (11)$$

$$\text{Platt: } \hat{p}' = \Lambda(a + b \text{ logit } \hat{p}), \quad (12)$$

$$\text{Isotonic: } \hat{p}' = \hat{m}(\hat{p}), \quad \hat{m} = \arg \min_{m \uparrow} \sum_i (D_i - m(\hat{p}_i))^2 \text{ (pool-adjacent-violators)}, \quad (13)$$

$$\text{Platt + grade: } \hat{p}' = \Lambda(a + b \text{ logit } \hat{p} + \sum_{g \neq A} \gamma_g \mathbf{1}\{\text{grade} = g\}), \quad (14)$$

with $\Lambda(x) = 1/(1 + e^{-x})$.

Pre-specified decision rule (2016 evidence only).

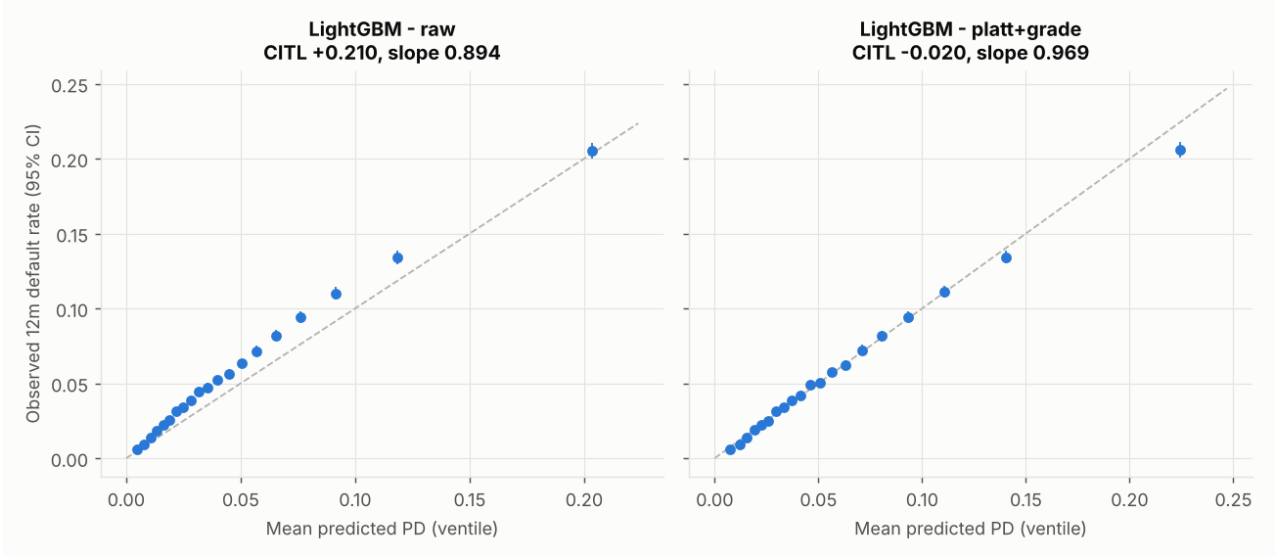
1. 2016 slope of the champion = 0.924, 95% CI [0.910, 0.938], $p(b = 1) = 3.2 \times 10^{-26} \Rightarrow$ Platt instead of intercept shift.
2. Residual grade effect after Platt: LR test Platt+grade vs Platt, $\chi^2 = 101.4$, $\text{df} = 6$, $p = 1.3 \times 10^{-19} \Rightarrow$ adopt Platt + grade (grades D and E were under-predicted in 2016).

Table 11: 2017 calibration of every model and method (raw models under-predict: mean PD $\approx 4.8\%$ vs DR 5.80%).

Model	Method	PD%	CITL	p	Slope	$p(=1)$	HL χ^2_8	Spieg. p	ECE pp	Brier
LogReg_WOE	raw	4.76	+0.215	$< 10^{-237}$	1.028	0.0039	1120.0	$< 10^{-232}$	1.04	0.0531
LogReg_WOE	intercept_shift	5.91	-0.021	0.0012	1.028	0.0039	36.4	0.0003	0.20	0.0529
LogReg_WOE	platt	5.89	-0.017	0.0079	0.977	0.0151	29.8	0.0147	0.17	0.0529
LogReg_WOE	isotonic	5.88	-0.016	0.0136	0.975	0.0071	23.4	0.0348	0.18	0.0529
LightGBM	raw	4.80	+0.210	$< 10^{-220}$	0.894	$< 10^{-47}$	1279.3	$< 10^{-259}$	1.00	0.0524
LightGBM	intercept_shift	5.88	-0.016	0.0136	0.894	$< 10^{-47}$	215.3	0.3639	0.40	0.0524
LightGBM	platt	5.91	-0.022	0.0011	0.968	$< 10^{-4}$	69.4	0.0486	0.20	0.0523
LightGBM	isotonic	5.91	-0.021	0.0013	0.958	$< 10^{-6}$	41.5	0.0221	0.23	0.0523
XGBoost	raw	4.70	+0.233	$< 10^{-271}$	0.917	$< 10^{-26}$	1397.1	$< 10^{-299}$	1.10	0.0525
XGBoost	intercept_shift	5.86	-0.012	0.0635	0.917	$< 10^{-26}$	116.6	0.6877	0.31	0.0524
XGBoost	platt	5.89	-0.018	0.0074	0.983	0.0380	30.0	0.0492	0.17	0.0523
XGBoost	isotonic	5.89	-0.018	0.0056	0.973	0.0010	25.1	0.0250	0.16	0.0523
RandomForest	raw	4.86	+0.192	$< 10^{-189}$	1.075	$< 10^{-14}$	933.0	$< 10^{-178}$	0.95	0.0527
RandomForest	intercept_shift	6.04	-0.045	$< 10^{-11}$	1.075	$< 10^{-14}$	103.3	$< 10^{-14}$	0.33	0.0526
RandomForest	platt	6.02	-0.041	$< 10^{-9}$	0.981	0.0250	57.7	$< 10^{-8}$	0.23	0.0526
RandomForest	isotonic	6.02	-0.041	$< 10^{-9}$	0.975	0.0044	54.9	$< 10^{-8}$	0.23	0.0526
LogReg_WOE_v2	raw	4.74	+0.220	$< 10^{-247}$	1.012	0.1954	1156.2	$< 10^{-244}$	1.06	0.0530
LogReg_WOE_v2	intercept_shift	5.89	-0.017	0.0080	1.012	0.1954	26.9	0.0051	0.15	0.0529
LogReg_WOE_v2	platt	5.88	-0.015	0.0231	0.980	0.0343	26.4	0.0410	0.15	0.0529
LogReg_WOE_v2	isotonic	5.87	-0.014	0.0302	0.976	0.0085	27.1	0.0722	0.19	0.0529
Stacked	raw	5.88	-0.016	0.0175	0.977	0.0034	34.1	0.1261	0.17	0.0523

Table 12: Final calibration on 2017 with 95% confidence intervals.

Model (2017, DR 5.80%)	Mean PD	CITL [95% CI]	Slope [95% CI]	HL χ^2_8	Spiegelhalter $z(p)$	ECE pp
LightGBM + Platt	5.91%	-0.0215 [-0.034, -0.009]	0.9677 [0.952, 0.983]	69.4	-1.97 (0.049)	0.20
LightGBM + Platt + grade (final)	5.90%	-0.0196 [-0.033, -0.007]	0.9692 [0.954, 0.985]	81.8	-1.77 (0.076)	0.24
LogReg v2 + Platt	5.88%	-0.0148 [-0.028, -0.002]	0.9804 [0.962, 0.999]	26.4	-2.04 (0.041)	0.15

**Figure 7:** Reliability diagram (20 ventiles, 95% binomial CI): raw LightGBM vs Platt + grade.

Interpretation. Recalibration moves mean PD from 4.80% to 5.90% (observed 5.80%), CITL from +0.210 to -0.020 (slightly conservative), ECE from 1.00 pp to 0.24 pp; Spiegelhalter $p = 0.076$ (no overall miscalibration at 5%). The residual slope $0.969 < 1$ means the extreme PDs are slightly too spread; the HL statistic (81.8) is driven by the top decile, where PD over-states the default rate (conservative).

7 Backtesting

For a bucket with N loans, D defaults and mean PD $\bar{p}$:

- **Exact binomial test** $H_0 : \text{DR} = \bar{p}$, $D \sim \text{Bin}(N, \bar{p})$, two-sided.
- **Jeffreys test** (ECB TRIM): posterior $\text{DR} \mid D \sim \text{Beta}(D + \frac{1}{2}, N - D + \frac{1}{2})$, $p = F_{\text{Beta}}(\bar{p}) = P(\text{DR} \leq \bar{p} \mid D)$. A small p means PD is too low.
- **Traffic light:** red if $p < 0.01$, amber if $p < 0.05$, “conservative” if $p > 0.99$, otherwise green.

Table 13: Backtest by grade, 2017 (final PD).

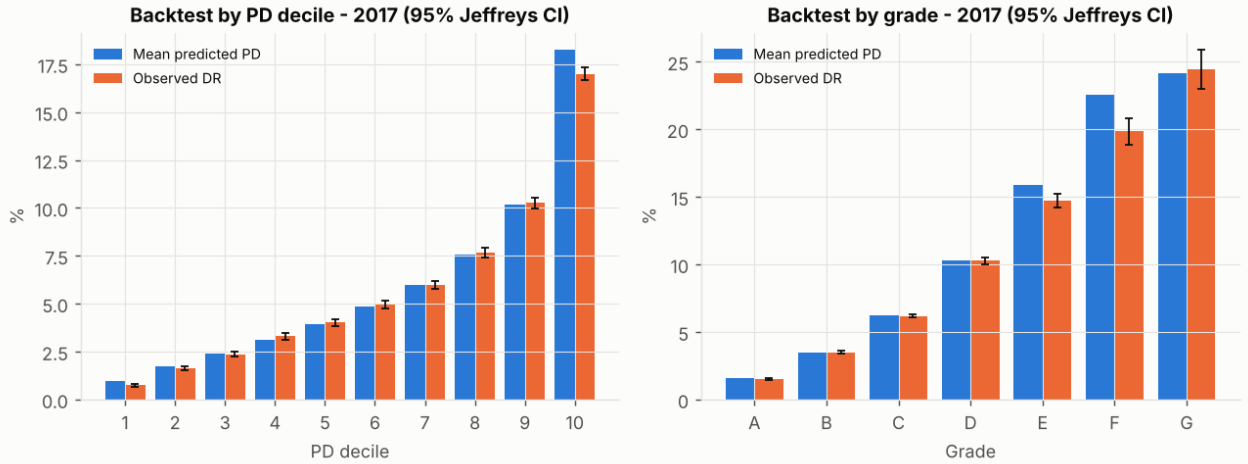
Grade	N	D	Mean PD	ODR	Binomial p	Jeffreys p	Light
A	78,796	1,202	1.59%	1.53%	0.1756	0.9153	green
B	133,127	4,670	3.50%	3.51%	0.9287	0.4647	green
C	145,144	8,978	6.20%	6.19%	0.8319	0.5855	green
D	56,646	5,816	10.27%	10.27%	0.9945	0.4975	green
E	20,167	2,971	15.89%	14.73%	$< 10^{-5}$	1.0000	conservative
F	6,242	1,239	22.55%	19.85%	$< 10^{-6}$	1.0000	conservative
G	3,457	844	24.17%	24.41%	0.7356	0.3664	green

Table 14: Backtest by PD decile, 2017.

Pd Decile	N	D	Mean PD	ODR	Binomial p	Jeffreys p	Light
1	44,358	330	0.97%	0.74%	$< 10^{-6}$	1.0000	conservative
2	44,358	723	1.73%	1.63%	0.0943	0.9553	green
3	44,358	1,052	2.42%	2.37%	0.5466	0.7305	green
4	44,358	1,466	3.13%	3.30%	0.0345	0.0174	amber (PD too low)
5	44,358	1,786	3.92%	4.03%	0.2401	0.1205	green
6	44,357	2,209	4.84%	4.98%	0.1701	0.0849	green
7	44,358	2,651	5.99%	5.98%	0.8965	0.5552	green
8	44,358	3,409	7.58%	7.69%	0.3941	0.1966	green
9	44,358	4,551	10.19%	10.26%	0.6099	0.3051	green
10	44,358	7,543	18.24%	17.00%	$< 10^{-10}$	1.0000	conservative

Table 15: Backtest by issue quarter, 2016 (in-sample for recalibration) and 2017 (out-of-sample).

Issue Q	N	D	Mean PD	ODR	Binomial p	Jeffreys p	Light
2016Q1	133,887	8,162	6.22%	6.10%	0.0611	0.9698	green
2016Q2	97,854	6,135	6.05%	6.27%	0.0045	0.0022	red (PD too low)
2016Q3	99,120	6,606	6.54%	6.66%	0.1000	0.0502	green
2016Q4	103,546	6,358	6.31%	6.14%	0.0249	0.9879	green
2017Q1	96,779	5,392	5.96%	5.57%	$< 10^{-6}$	1.0000	conservative
2017Q2	105,451	6,132	5.91%	5.82%	0.2170	0.8928	green
2017Q3	122,701	7,377	6.13%	6.01%	0.0782	0.9614	green
2017Q4	118,648	6,819	5.61%	5.75%	0.0362	0.0181	amber (PD too low)

**Figure 8:** Predicted PD vs observed default rate with 95% Jeffreys intervals.

Result. Grades: 5 green, 2 conservative, no red or amber (grade D was red under Platt alone: 9.76% vs 10.27%). Deciles: 7 green, 2 conservative, 1 amber (decile 4: 3.13% vs 3.30%).

8 Stability and vintage analysis

Population stability index. With expected (development) shares e_k and actual shares a_k over 10 score deciles of the development sample plus a missing bin,

$$\text{PSI} = \sum_k (a_k - e_k) \ln \frac{a_k}{e_k}; \quad < 0.10 \text{ stable, } 0.10\text{--}0.25 \text{ monitor, } > 0.25 \text{ significant shift.} \quad (15)$$

The characteristic stability index (CSI) is the same formula applied to one feature.

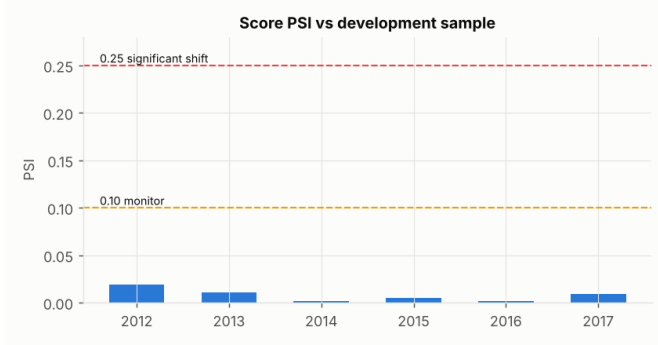


Figure 9: Score PSI against the development sample: stable in every year.

Table 16: CSI of the top numeric features. Large values against the full development window come from the missing bin (fields structurally missing before 2012); against a fully populated 2013–2015 reference they disappear. `int_rate` and `bc_util` show genuine moderate drift.

Feature	vs dev 2016	vs dev 2017	vs 2013–15: 2016	vs 2013–15: 2017	Dev missing
sub_grade_n	0.021	0.045	0.022	0.045	0.0%
int_rate	0.091	0.144	0.091	0.148	0.0%
acc_open_past_24mths	0.595	0.587	0.009	0.002	5.3%
dti	0.007	0.022	0.003	0.020	0.0%
inst_to_inc	0.018	0.053	0.023	0.062	0.0%
revol_to_inc	0.015	0.034	0.018	0.038	0.0%
mo_sin_old_rev_tl_op	0.883	0.919	0.020	0.058	7.6%
log_inc	0.009	0.018	0.006	0.016	0.0%
bc_util	0.128	0.182	0.035	0.104	6.3%
mo_sin_old_il_acct	0.119	0.120	0.006	0.019	10.5%
loan_to_inc	0.015	0.044	0.021	0.052	0.0%
mths_since_recent_inq	0.020	0.012	0.001	0.004	15.1%
mths_since_recent_bc	0.102	0.086	0.006	0.004	6.2%
credit_hist_m	0.004	0.006	0.003	0.005	0.0%
bc_open_to_buy	0.119	0.161	0.024	0.081	6.2%
emp_len	0.011	0.018	0.011	0.019	5.1%
revol_util	0.035	0.094	0.037	0.100	0.1%
installment	0.015	0.033	0.020	0.041	0.0%

Vintage analysis. For issue year v , the cumulative default rate at month-on-book m is $\text{CDR}_v(m) = \frac{1}{N_v} \sum_{i \in v} \mathbf{1}\{\text{status}_i \in \mathcal{B}, \tau_i \leq m\}$, shown only for fully observed months.

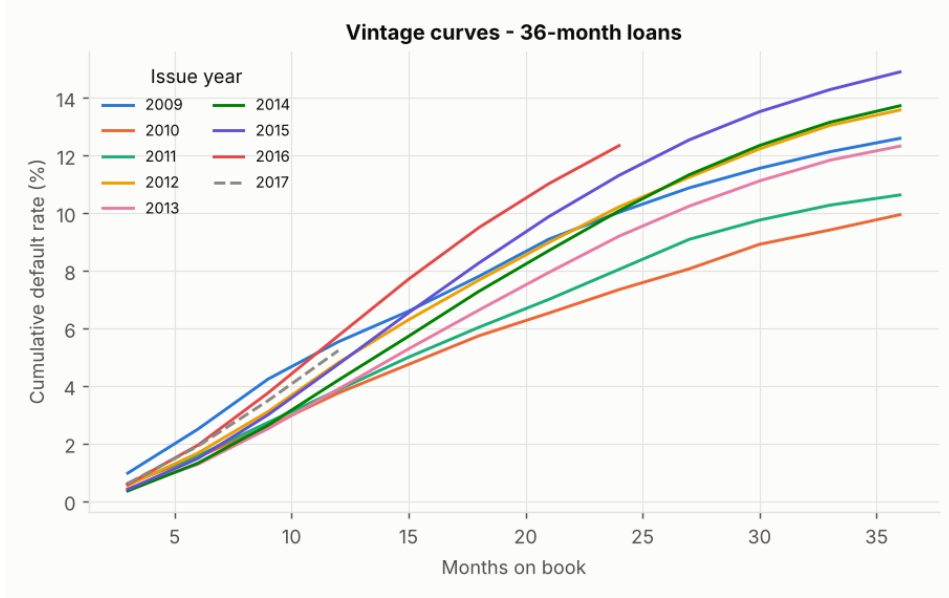


Figure 10: Vintage curves, 36-month loans. Deterioration from 2013 to 2016: CDR at 24 months 9.2% (2013), 10.1% (2014), 11.3% (2015), 12.3% (2016).

9 Loss given default and exposure at default

9.1 LGD

For a defaulted loan,

$$\text{EAD}_i = \text{Funded}_i - \text{PrincipalRepaid}_i, \quad \text{LGD}_i = 1 - \frac{\max(\text{Recoveries}_i - \text{CollectionFee}_i, 0)}{\text{EAD}_i} \in [0, 1]. \quad (16)$$

Sample. Recoveries flatten about 18 months after default, so only charge-offs at least 18 months before the snapshot are used (173,643 loans). Mean LGD: 0.897 (18–36 months since default) vs 0.889 (37–72 months), Welch $p \approx 10^{-52}$ but only 0.9 pp apart. OOT split by default date: train < July 2015 ($n = 36,761$), test July 2015 – September 2017 ($n = 136,882$).

Fractional logit (Papke–Wooldridge): $\mathbb{E}[\text{LGD} \mid \mathbf{x}] = \Lambda(\mathbf{x}'\boldsymbol{\beta})$, estimated by Bernoulli quasi-maximum likelihood $\max_{\boldsymbol{\beta}} \sum_i [y_i \ln \Lambda(\mathbf{x}'_i \boldsymbol{\beta}) + (1 - y_i) \ln(1 - \Lambda(\mathbf{x}'_i \boldsymbol{\beta}))]$ with HC1 robust standard errors (consistent for $y \in [0, 1]$).

Diebold–Mariano style test against the mean benchmark: $d_i = (y_i - \hat{y}_i^{\text{bench}})^2 - (y_i - \hat{y}_i^{\text{model}})^2$, $t = \bar{d}/(s_d/\sqrt{n})$, one-sided $H_1 : \mathbb{E}d > 0$.

Table 17: LGD models, out-of-time.

Model	Mean obs.	Mean pred.	RMSE	MAE	R^2_{OOS}	Spearman	DM t vs mean	p (1-sided)
mean	0.8954	0.8954	0.1169	0.0735	-0.0000	–	–	–
segment_grade_term	0.8954	0.8952	0.1170	0.0736	-0.0023	0.049	-8.38	1.0000
frac_logit_glm	0.8954	0.8899	0.1168	0.0755	0.0015	0.115	1.62	0.0522
lightgbm_xentropy	0.8954	0.8877	0.1173	0.0771	-0.0072	0.096	-6.58	1.0000

Decision. No borrower-level model beats the mean out-of-time (GLM $p = 0.052$; segment and GBM are significantly *worse*). LendingClub recoveries come mostly from bulk debt sales, which are not driven by borrower characteristics. Hence $\text{LGD} = \overline{\text{LGD}} = \mathbf{0.8954}$ (long-run, default-weighted). **Downturn LGD** = $\max_y \overline{\text{LGD}}_y$ over default years with $\geq 1,000$ defaults = **0.9363** (2012).

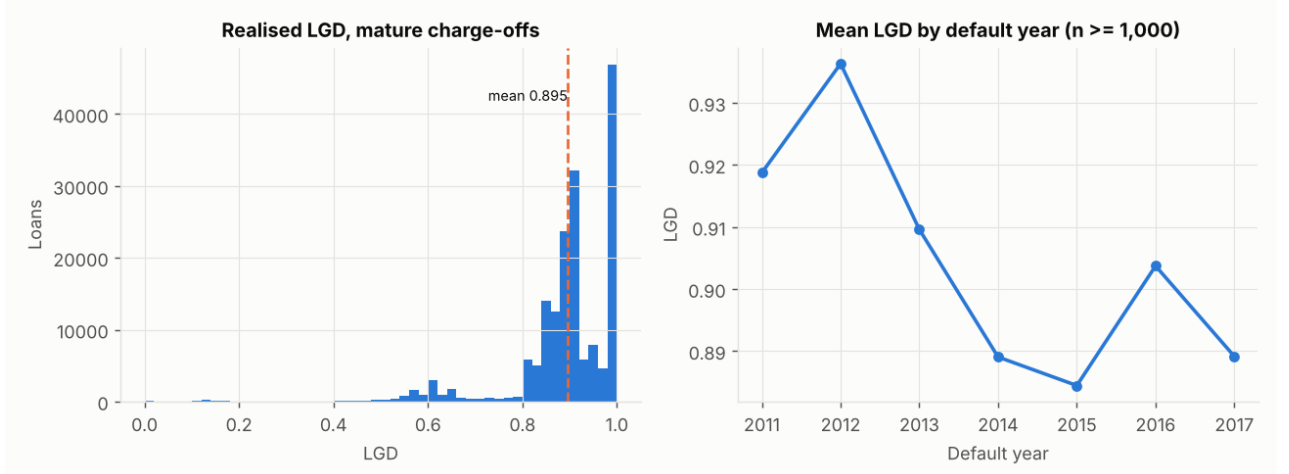


Figure 11: Distribution of realised LGD and mean LGD by default year.

9.2 EAD

For loans defaulting within 12 months, the EAD ratio $r_i = \text{EAD}_i / \text{Funded}_i$ (outstanding principal share at default) is modelled with a fractional logit using term, interest rate, log amount, FICO, DTI, revolving utilisation, log income, home ownership and purpose. OOT: train issue years 2007–2015, test 2016 ($n = 27,261$).

Table 18: EAD-ratio models, out-of-time.

Model	Mean obs.	Mean pred.	RMSE	MAE	R_{oos}^2	Spearman	DM t vs mean	p (1-sided)
mean	0.8731	0.8744	0.0762	0.0632	-0.0003	–	–	–
segment_term	0.8731	0.8685	0.0662	0.0517	0.2447	0.521	50.14	$< 10^{-299}$
frac_logit_glm	0.8731	0.8707	0.0642	0.0497	0.2912	0.542	54.43	$< 10^{-299}$

The GLM is clearly better than the mean ($R_{oos}^2 = 0.29$, DM $t = 54.4$). Longer terms and higher rates mean slower amortisation and therefore higher EAD.

10 Expected credit loss

10.1 Formulas

$$\text{ECL}^{12m} = \sum_i \text{PD}_i^{12m} \cdot \text{LGD} \cdot \text{EAD}_i^{12m}, \quad \text{EAD}_i^{12m} = \hat{r}_i \cdot \text{Funded}_i. \quad (17)$$

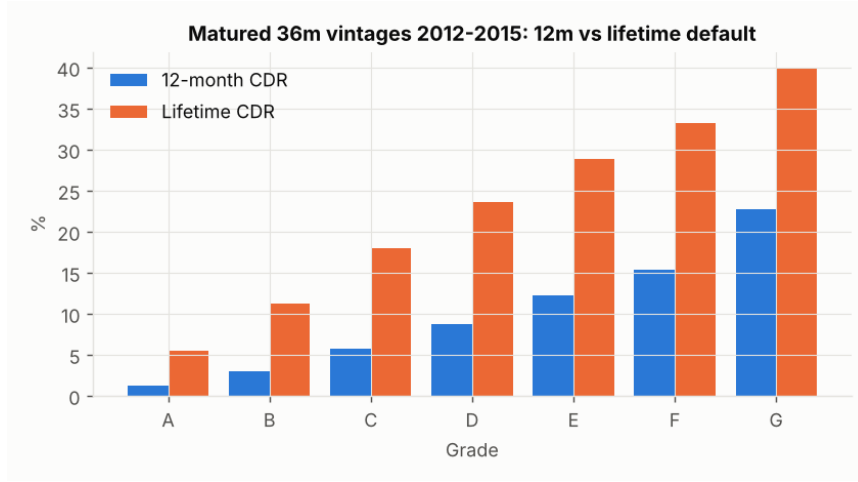
Lifetime PD. Using the cumulative hazard $H = -\ln(1 - \text{PD})$ and assuming a proportional hazard term structure within grade g and term t , estimated on fully matured vintages (36m: 2012–2015, 60m: 2012–2013):

$$k_{g,t} = \frac{\ln(1 - \text{CDR}_{g,t}^{life})}{\ln(1 - \text{CDR}_{g,t}^{12m})}, \quad \text{PD}_i^{life} = 1 - (1 - \text{PD}_i^{12m})^{k_{g,t}}, \quad (18)$$

$$\text{ECL}^{life} = \sum_i \text{PD}_i^{life} \cdot \text{LGD} \cdot \bar{r}_{g,t}^{life} \text{Funded}_i. \quad (19)$$

Table 19: PD term structure from matured vintages.

Grade	Term	n	CDR_{12}	CDR_{life}	k	EAD ratio (life)
A	36	133,275	1.26%	5.45%	4.409	0.507
A	60	770	1.82%	8.31%	4.729	0.606
B	36	202,361	2.97%	11.23%	3.952	0.539
B	60	5,505	2.92%	15.11%	5.520	0.632
C	36	156,094	5.76%	18.04%	3.354	0.581
C	60	15,414	4.21%	21.73%	5.697	0.653
D	36	72,843	8.75%	23.57%	2.934	0.621
D	60	8,299	6.75%	27.53%	4.610	0.699
E	36	20,542	12.26%	28.92%	2.609	0.655
E	60	8,218	8.31%	32.04%	4.451	0.723
F	36	4,054	15.34%	33.18%	2.420	0.685
F	60	4,997	10.83%	36.30%	3.936	0.742
G	36	466	22.75%	39.91%	1.974	0.736
G	60	1,086	12.34%	37.94%	3.622	0.762

**Figure 12:** 12-month vs lifetime cumulative default rate, matured 36-month vintages.

10.2 Results for the 2017 cohort

Table 20: ECL by grade, 2017 cohort (443,579 loans, \$6,585.0M funded; LGD = 0.8954).

Grade	Loans	Funded \$M	PD_{12}	ODR_{12}	PD_{life}	ECL_{12} \$M	%	ECL_{life} \$M	%
A	78,796	1,096.9	1.59%	1.53%	6.77%	12.4	1.13%	33.0	3.01%
B	133,127	1,839.7	3.50%	3.51%	13.73%	48.0	2.61%	130.4	7.09%
C	145,144	2,224.9	6.20%	6.19%	22.79%	105.4	4.74%	293.2	13.18%
D	56,646	891.9	10.27%	10.27%	30.95%	73.4	8.23%	170.9	19.16%
E	20,167	340.7	15.89%	14.73%	43.82%	45.1	13.24%	98.6	28.94%
F	6,242	118.6	22.55%	19.85%	57.52%	22.5	18.98%	45.8	38.61%
G	3,457	72.3	24.17%	24.41%	57.14%	14.8	20.42%	28.5	39.35%
Total	443,579	6,585.0	5.90%	5.80%	19.98%	321.5	4.88%	800.4	12.15%

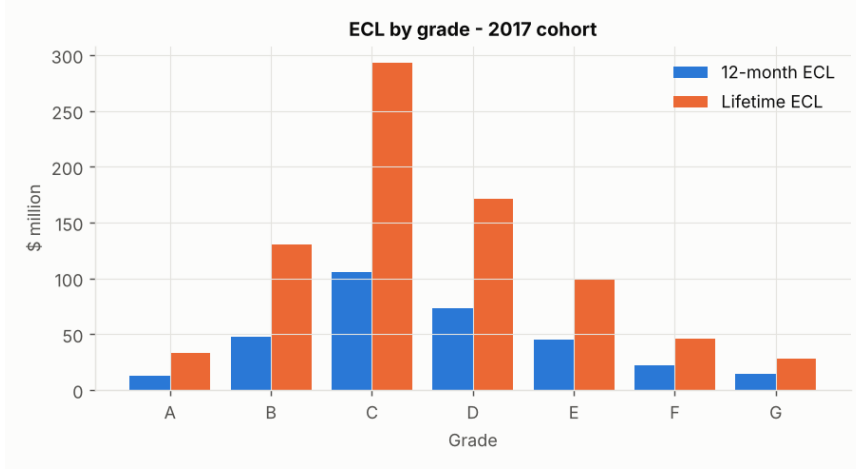


Figure 13: 12-month and lifetime ECL by grade.

ECL results and backtest

- 12-month EAD = \$5,719.0M; $ECL^{12m} = \$321.5M$ (4.88% of funded).
- $ECL^{life} = \$800.4M$ (12.15%), mean lifetime PD 19.98%.
- **Backtest:** realised 12-month loss = $\sum_{i:D_i=1} EAD_i \cdot LGD = \$326.1M$; ratio realised/predicted = 1.014. Realised EAD of defaulters \$364.2M vs predicted $\sum PD_i EAD_i = \$359.1M$.
- An independence-only test $z = (L^{real} - ECL) / \sqrt{\sum (LGD EAD_i)^2 PD_i (1 - PD_i)} = 2.08$ ($p = 0.04$) ignores default correlation; under the correlated loss distribution of Section 11 the realised loss is at the 59th percentile ($p = 0.82$).

11 Portfolio credit risk: one-factor Gaussian model

11.1 Model

Each borrower has a latent asset value driven by one systematic factor $Z \sim N(0, 1)$ and an idiosyncratic shock $\varepsilon_i \sim N(0, 1)$, independent:

$$A_i = -\sqrt{\rho_i} Z + \sqrt{1 - \rho_i} \varepsilon_i, \quad \text{default}_i \iff A_i < \Phi^{-1}(PD_i). \quad (20)$$

($Z > 0$ is a bad state.) Conditional on $Z = z$ defaults are independent with

$$PD_i(z) = \Phi\left(\frac{\Phi^{-1}(PD_i) + \sqrt{\rho_i} z}{\sqrt{1 - \rho_i}}\right), \quad \mathbb{E}_Z[PD_i(Z)] = PD_i. \quad (21)$$

Portfolio loss and risk measures:

$$L = \sum_{i=1}^N EAD_i \cdot LGD_i \cdot \mathbf{1}\{\text{default}_i\}, \quad EL = \mathbb{E}[L] = \sum_i PD_i LGD_i EAD_i, \quad (22)$$

$$\text{VaR}_q(L) = \inf\{\ell : P(L \leq \ell) \geq q\}, \quad \text{ES}_q(L) = \mathbb{E}[L \mid L \geq \text{VaR}_q(L)], \quad \text{UL}_q = \text{VaR}_q - EL. \quad (23)$$

ASRF benchmark. For a large granular portfolio, idiosyncratic risk diversifies away and

$$\text{VaR}_q(L) \approx \sum_i EAD_i LGD_i \Phi\left(\frac{\Phi^{-1}(PD_i) + \sqrt{\rho_i} \Phi^{-1}(q)}{\sqrt{1 - \rho_i}}\right). \quad (24)$$

11.2 Asset correlation

Vasicek distribution. For a large homogeneous cohort the default rate X has density

$$f(x; p, \rho) = \sqrt{\frac{1-\rho}{\rho}} \exp\left(\frac{1}{2}[\Phi^{-1}(x)]^2 - \frac{1}{2\rho}\left(\sqrt{1-\rho}\Phi^{-1}(x) - \Phi^{-1}(p)\right)^2\right). \quad (25)$$

Estimation. 36 quarterly cohorts (2009Q1–2017Q4, $n \geq 500$) with observed 12-month default rate DR_t and mean model PD $\bar{p}_t$ (risk mix). A free through-the-cycle level c prevents a level bias of the PDs from being read as systematic risk:

$$(\hat{\rho}, \hat{c}) = \arg \max_{\rho, c} \sum_t \ln f(DR_t; \Lambda(\text{logit } \bar{p}_t + c), \rho), \quad \hat{z}_t = \frac{\sqrt{1-\hat{\rho}}\Phi^{-1}(DR_t) - \Phi^{-1}(p_t)}{\sqrt{\hat{\rho}}}. \quad (26)$$

ρ is profiled over c ; the Wald SE comes from the curvature of the profile log-likelihood. Because the 12-month windows of adjacent quarters overlap, a moving-block bootstrap (block length 4, 1,000 replications) gives a robust CI. Without the free level, the implied $\hat{z}_t$ had mean -0.75 (biased); a unit test confirms that the free-level estimator recovers the true ρ when PDs are biased.

- Model-adjusted MLE: $\hat{\rho} = \mathbf{0.0037}$, Wald 95% CI $[0.0020, 0.0054]$, block bootstrap $[0.0016, 0.0048]$; $\hat{c} = -0.185$.
- Grade-pooled unconditional MLE: $\hat{\rho} = 0.0164$ $[0.0130, 0.0207]$ (includes underwriting drift).
- Basel IRB “other retail”: $\rho(\text{PD}) = 0.03w + 0.16(1-w)$, $w = \frac{1 - e^{-35\text{PD}}}{1 - e^{-35}}$; 0.046 at the mean PD, portfolio mean 0.065.

Choice. 2009–2017 contains no recession, so the data estimate measures benign-period co-movement and is a lower bound for tail risk. The **base case uses the Basel retail $\rho(\text{PD}_i)$ per loan**; the data estimate is a sensitivity.

Table 21: Quarterly cohorts used for the correlation estimate (DR and PD in %, $\hat{z}_t$ implied factor).

Quarter	n	DR%	PD%	$\hat{z}_t$	Quarter	n	DR%	PD%	$\hat{z}_t$
2009Q1	775	5.16	7.00	-1.02	2013Q3	37,571	4.47	5.99	-0.89
2009Q2	965	5.28	6.40	-0.09	2013Q4	43,869	4.21	5.82	-1.10
2009Q3	1,231	4.55	5.97	-0.72	2014Q1	47,410	4.25	5.98	-1.25
2009Q4	1,745	6.53	6.42	+1.63	2014Q2	55,349	4.59	6.22	-0.98
2010Q1	1,953	4.35	5.87	-0.92	2014Q3	58,726	4.94	6.50	-0.76
2010Q2	2,776	5.04	6.03	+0.01	2014Q4	74,144	5.04	6.47	-0.55
2010Q3	3,284	4.11	6.15	-1.73	2015Q1	84,277	5.21	6.44	-0.26
2010Q4	3,523	3.89	5.57	-1.36	2015Q2	95,825	5.38	6.55	-0.14
2011Q1	4,126	4.22	5.93	-1.24	2015Q3	110,489	5.52	6.48	+0.16
2011Q2	5,102	4.82	6.12	-0.46	2015Q4	130,504	5.79	6.30	+0.78
2011Q3	5,876	4.95	5.87	+0.09	2016Q1	133,887	6.10	6.22	+1.30
2011Q4	6,617	5.24	6.09	+0.25	2016Q2	97,854	6.27	6.05	+1.76
2012Q1	8,076	5.44	5.84	+0.87	2016Q3	99,120	6.66	6.54	+1.65
2012Q2	10,447	5.42	6.56	-0.09	2016Q4	103,546	6.14	6.31	+1.25
2012Q3	16,133	5.41	6.38	+0.11	2017Q1	96,779	5.57	5.96	+0.91
2012Q4	18,711	4.96	5.86	+0.11	2017Q2	105,451	5.82	5.91	+1.33
2013Q1	22,706	4.33	5.78	-0.84	2017Q3	122,701	6.01	6.13	+1.30
2013Q2	30,668	4.65	6.14	-0.77	2017Q4	118,648	5.75	5.61	+1.65

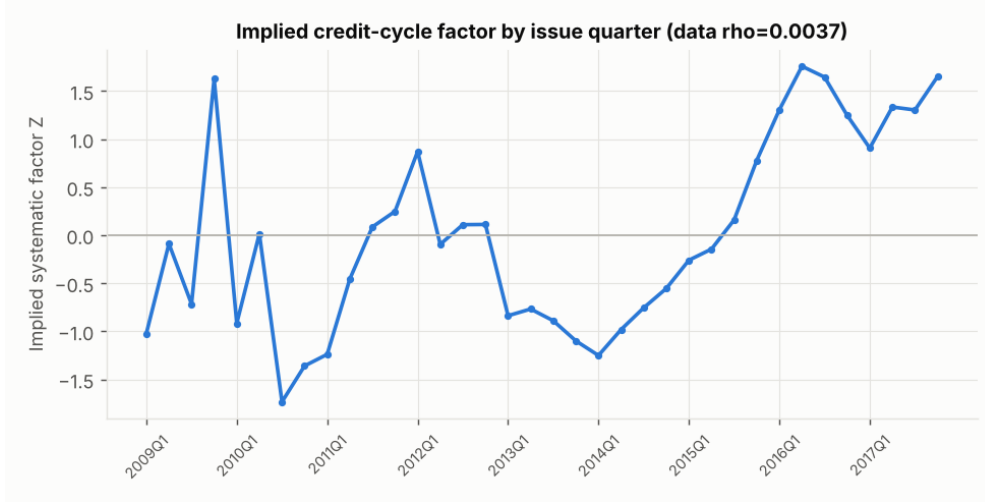


Figure 14: Implied systematic factor by issue quarter; 2016Q2 is the worst quarter ($\hat{z} = 1.76$).

11.3 Monte Carlo simulation

Algorithm 1 Loan-level Monte Carlo (all 443,579 loans, $S = 10,000$ scenarios)

- 1: Draw $z_1, \dots, z_S \stackrel{iid}{\sim} N(0, 1)$ ▷ common random numbers across stress scenarios
 - 2: **for** $s = 1, \dots, S$ **do**
 - 3: $p_{is} \leftarrow \Phi((\Phi^{-1}(\text{PD}_i) + \sqrt{\rho_i} z_s) / \sqrt{1 - \rho_i})$ for every loan i
 - 4: Draw $u_{is} \sim U(0, 1)$; $L_s \leftarrow \sum_i \text{EAD}_i \text{LGD}_i \mathbf{1}\{u_{is} < p_{is}\}$
 - 5: **end for**
 - 6: $\widehat{\text{EL}} = \bar{L}$; $\widehat{\text{VaR}}_q = L_{(\lceil qS \rceil)}$; $\widehat{\text{ES}}_q = \text{mean}\{L_s : L_s \geq \widehat{\text{VaR}}_q\}$
-

Uncertainty and validity checks.

- **VaR CI** (distribution-free): $[L_{(j)}, L_{(k)}]$ with $j = F_{\text{Bin}(S,q)}^{-1}(0.025)$, $k = F_{\text{Bin}(S,q)}^{-1}(0.975)$. **ES CI**: $\widehat{\text{ES}} \pm 1.96 s_{\text{tail}} / \sqrt{n_{\text{tail}}}$.
- **EL check**: $z = (\bar{L} - \text{EL}) / (s_L / \sqrt{S}) = -0.77$, $p = 0.44$.
- **ASRF check**: MC $\text{VaR}_{99\%}$ is +0.96% and $\text{VaR}_{99.9\%}$ +0.69% above the analytic ASRF values (\$716.8M, \$933.4M); the small excess is the granularity add-on.
- **Original set-up** (10,000 loans $\times$ 2,000 scenarios): 99% VaR CI width = 16.4% of VaR, i.e. about $\pm 8\%$ Monte Carlo error.

Table 22: Convergence of 99% VaR with the number of scenarios (base case).

Scenarios	VaR _{99%} \$M	95% CI \$M	CI width
500	670.2	[655.3, 880.8]	33.7%
1,000	670.0	[652.1, 786.1]	20.0%
2,000	706.0	[665.2, 750.0]	12.0%
5,000	726.0	[700.6, 751.0]	6.9%
10,000	723.7	[703.3, 747.3]	6.1%

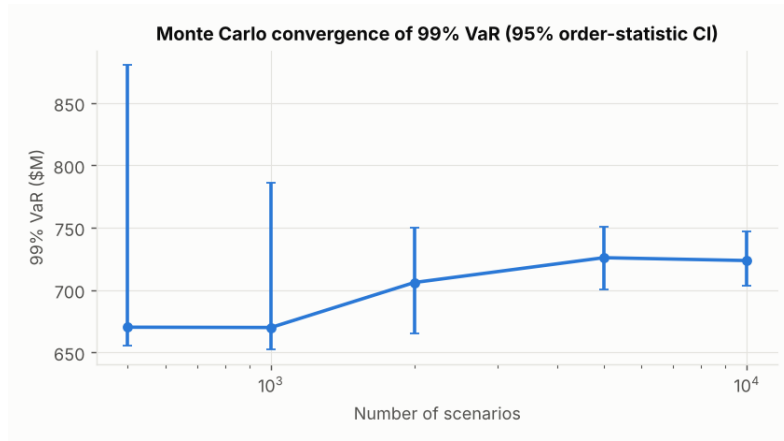


Figure 15: Convergence of the 99% VaR estimate.

11.4 Base-case results

Table 23: Base case: 2017 portfolio, 12-month horizon, Basel retail correlation, LGD 0.895.

Measure	Value (\$M)	95% CI (\$M)
Expected loss (analytic / MC)	321.5 / 320.5	MC SE 1.3
Standard deviation of loss	128.0	
99% VaR	723.7	[703.3, 747.3]
99% ES	820.6	[802.6, 838.5]
99.9% VaR	939.8	[916.0, 994.0]
99.9% ES	1,020.0	[964.3, 1,075.6]
Unexpected loss (99.9% VaR – EL)	618.3	9.4% of funded

Loss backtest. Realised 2017 12-month loss \$326.1M lies at the **58.9th percentile** of the simulated distribution: $p = 2 \min(\hat{F}(L^{real}), 1 - \hat{F}(L^{real})) = 0.82$, so the model is not rejected.

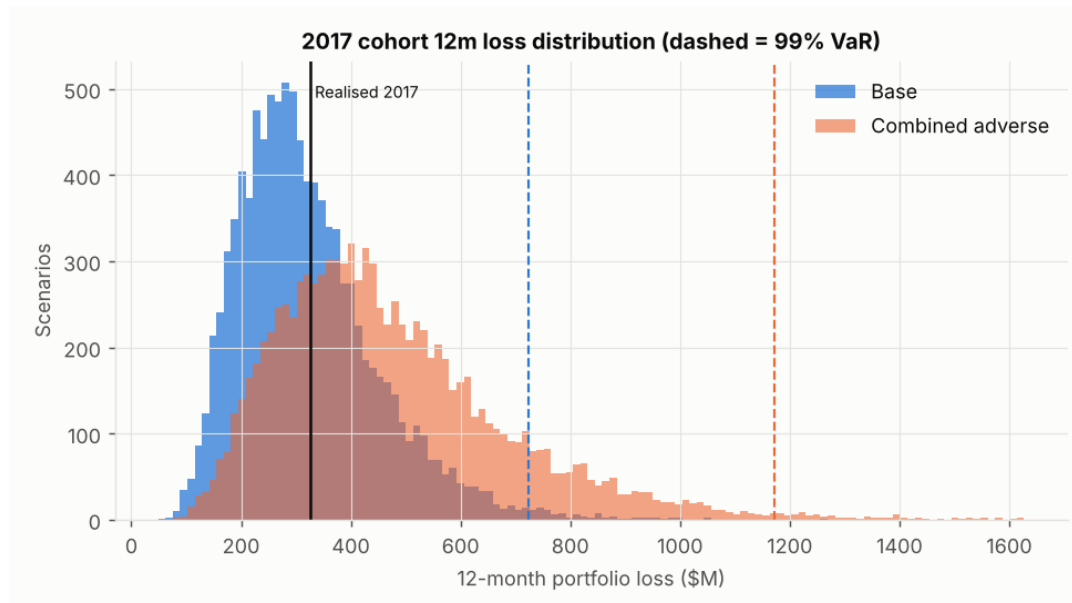


Figure 16: Simulated 12-month loss distribution: base vs combined adverse scenario, with the realised 2017 loss.

12 Stress testing

All scenarios reuse the same random draws (common random numbers), so differences reflect the scenario rather than Monte Carlo noise.

- **PD odds stress:** $PD'_i = \Lambda(\text{logit } PD_i + \ln m)$, $m \in \{1.5, 2\}$. An odds multiplier keeps PDs in $(0, 1)$ and stresses low and high PDs proportionally in odds.
- **Historical scenario** (empirical): $m_{hist} = \max_t \frac{\text{odds}(DR_t)}{\text{odds}(\Lambda(\text{logit } \bar{p}_t + \hat{c}))} = 1.249$ (2016Q2).
- **LGD stress:** downturn LGD 0.936; LGD +5 pp.
- **Correlation stress:** $\rho'_i = \min(1.5\rho_i, 0.99)$; sensitivity with the data-estimated $\rho = 0.0037$.
- **Combined adverse:** PD odds $\times 1.5$ + downturn LGD + $\rho \times 1.5$.
- **Reverse stress:** find m^* such that $\sum_i \Lambda(\text{logit } PD_i + \ln m^*) \text{LGD EAD}_i = \text{VaR}_{99.9\%}^{base}$ (Brent's method): $m^* = 3.74$.

Table 24: Stress-test results (\$M).

Scenario	EL	VaR _{99%}	ES _{99%}	VaR _{99.9%}	ES _{99.9%}
Base	320.5	723.7	820.6	939.8	1,020.0
PD odds x1.5	455.4	966.7	1,082.5	1,226.4	1,317.0
PD odds x2.0	578.2	1,174.4	1,303.1	1,462.3	1,560.8
Historical worst quarter 2016Q2 (PD odds x1.25)	389.4	850.9	957.3	1,090.7	1,175.5
Downturn LGD (0.936)	335.2	756.8	858.1	982.8	1,066.6
LGD +5pp	338.4	764.1	866.4	992.3	1,076.9
Correlation x1.5	320.4	850.8	988.7	1,160.4	1,276.2
Data-estimated rho (0.0037, benign 2009-17)	321.2	411.2	426.2	446.3	456.2
Combined adverse: PD x1.5 + downturn LGD + rho x1.5	476.0	1,172.3	1,343.1	1,550.8	1,687.6

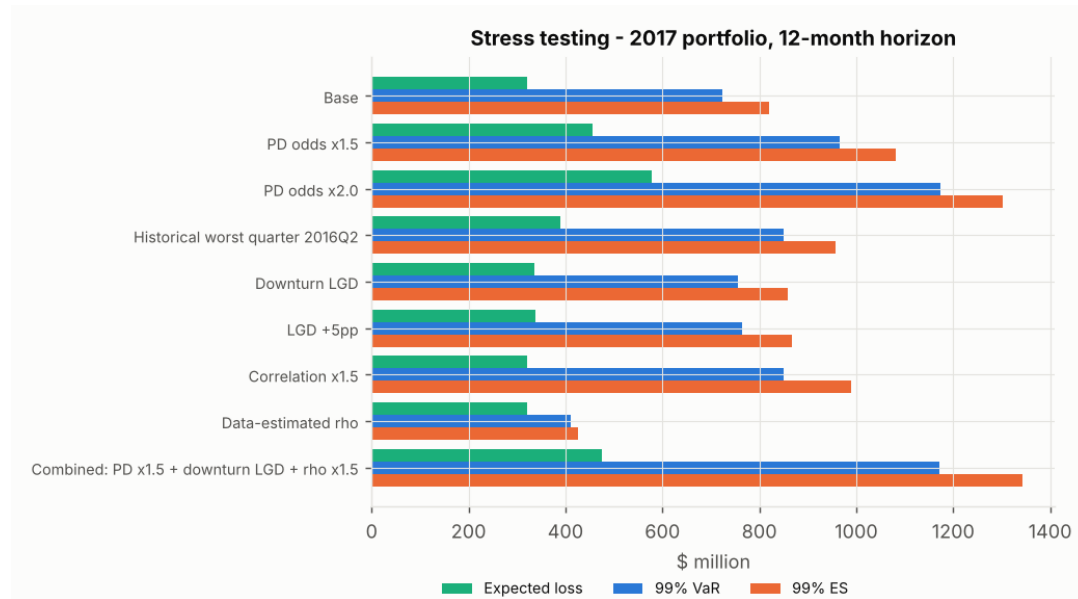


Figure 17: Expected loss, 99% VaR and 99% ES by scenario.

Interpretation. Tail risk is driven mainly by PD level and correlation: PD odds $\times 2$ raises 99% VaR by 62%; $\rho \times 1.5$ raises it by 18% with unchanged EL. LGD stresses move it by only about 5% because LGD is already close to 0.9. The data-estimated correlation would cut 99% VaR to \$411M, which is why it is not used for capital.

13 Comparison with the original project

Table 25: Original claims vs this study.

Item	Original	This study (statistically verified)
Sample / target	1.35M terminal loans, lifetime default	1.76M loans, 12-month default, no survivorship bias
2017 cohort	169,321 loans, \$2.42bn	443,579 loans, \$6.585bn
PD discrimination	LR AUC 0.700, KS 0.291, Gini 0.400	LightGBM AUC 0.725, KS 0.325, Gini 0.450 (DeLong vs LR $p < 10^{-100}$)
Calibration	not tested	Platt + grade; CITL -0.02 , ECE 0.24 pp, Jeffreys no red grade
LGD	assumption	empirical 0.895, downturn 0.936; OOT model tests
ECL	\$497.3M on \$2.42bn	12m \$321.5M (backtest ratio 1.014); lifetime \$800.4M
Monte Carlo	10k loans, 2k scenarios	443,579 loans, 10k scenarios, CIs, ASRF check
99% VaR / ES	\$71.4M / \$78.3M (10k-loan sample)	\$723.7M / \$820.6M (full portfolio)
Stress	PD odds, LGD	+ historical, correlation, combined, reverse stress

14 Limitations

- Public data only: no macro-economic covariates, no collections or recovery-process data; LGD reflects LendingClub’s debt-sale recoveries.
- The 12-month default date is reconstructed from payment information (98% agreement with the date rule).
- Lifetime PD extrapolates 12-month PD with hazard ratios from 2012–2015 vintages; 2017 lifetime outcomes are not yet observable.
- The base correlation is a supervisory calibration; the observed window has no recession.
- Hosmer–Lemeshow rejects at $N = 443k$ because of its high power; effect sizes are small.

Reproducibility

Code: `src/crm` (library) and `scripts/01-09` (pipeline; `run_all.sh`). 19 unit tests validate the statistical machinery: DeLong AUC vs scikit-learn, DeLong variance vs bootstrap, DeLong test size under H_0 , KS vs SciPy, calibration tests on perfect and miscalibrated simulated models, PSI, Jeffreys, Holm, WOE monotonicity, Vasicek density integrating to 1, $\mathbb{E}[\text{PD}(Z)] = \text{PD}$, 95% CI coverage of the ρ MLE, free-level estimator under biased PDs, MC EL vs analytic EL, and MC VaR vs ASRF (scalar and per-loan ρ).

15 Interview questions and model answers

A. Data, target and sample design

Q1. Why did you not use only Fully Paid and Charged Off loans?

Answer. For recent cohorts only loans that already ended are observed, and those are mostly early defaults and early prepayments. In 2017 only 38% of loans had ended and their 12-month default rate was 15.1% vs 5.8% for all loans. That is survivorship (selection) bias. A fixed-horizon target defined for every loan removes it.

Q2. How did you define default and why a 12-month horizon?

Answer. Charge-off, default or 31–120 days late with the first missed instalment within 12 months on book. 12 months is the Basel PD horizon and the IFRS 9 stage-1 horizon, and it matches the 1-year horizon of the Vasicek portfolio model.

Q3. How did you reconstruct the default date?

Answer. From the last payment date when no recovery cash was received, otherwise from the number of instalments paid ($\lceil (P + Int)/A \rceil + 1$). I validated the rule on 104k clean charge-offs: 98% agreement on the 12-month flag.

Q4. What is out-of-time validation and why is it better than a random split?

Answer. Train on earlier vintages and test on later ones. It mimics deployment and reveals drift in the population and in default rates; a random split would leak the 2016–2017 environment into training and overstate performance.

Q5. How did you avoid data leakage?

Answer. Only origination-time fields are inputs. Payment, recovery, last-FICO, hardship and settlement fields are used only for targets, LGD and EAD. All decisions use data up to 2016; 2017 is scored once.

B. Feature engineering and selection

Q6. Explain WOE and IV. What thresholds do you use?

Answer. $WOE_b = \ln(\%bad_b / \%good_b)$; $IV = \sum (\%bad - \%good) WOE$. Rule of thumb: < 0.02 useless, 0.02 – 0.1 weak, 0.1 – 0.3 medium, > 0.3 strong (beware > 0.5 : possible leakage). I used $IV \geq 0.02$.

Q7. Why monotone binning?

Answer. It gives a logical, explainable relationship (risk always increases in one direction), avoids over-fitting noise in sparse bins and is expected by validators and regulators.

Q8. How did you handle multicollinearity?

Answer. A $|r| > 0.7$ filter in IV order before selection and a $VIF < 5$ gate during selection. Example: `int_rate` was dropped because it is almost a function of `sub_grade`.

Q9. Why does the scorecard select only 5–12 variables?

Answer. Each variable must add holdout AUC, be significant at 0.1% and keep the right sign. LendingClub's sub-grade already prices FICO and the bureau data (FICO has Wald $p \approx 0.3$ once sub-grade is in), so few variables add independent information.

Q10. How do you compare two nested logistic models?

Answer. Likelihood-ratio test $2(\ell_1 - \ell_0) \sim \chi^2_{\Delta k}$, plus AIC/BIC and a DeLong test on out-of-time data. v2 vs v1: $\chi^2 = 793$ on 7 df, $p \approx 10^{-167}$, but only +0.001 AUC.

C. Models and discrimination

Q11. Why is LightGBM better than logistic regression here?

Answer. It captures non-linearities and interactions (e.g. state $\times$ grade, utilisation $\times$ income) that a WOE-additive model cannot. The gain is +0.023 AUC, highly significant by DeLong.

Q12. What is the DeLong test and why use it instead of comparing two confidence intervals?

Answer. It estimates the variance and the covariance of correlated AUCs computed on the same loans; the paired test is much more powerful than checking whether separate CIs overlap.

Q13. Relationship between AUC, Gini and KS?

Answer. $\text{Gini} = 2\text{AUC} - 1$. KS is the maximum vertical distance between the cumulative score distributions of defaulters and non-defaulters; it is a single-cut-off measure while AUC averages over all cut-offs.

Q14. Why not deploy the stacked model?

Answer. It adds +0.0009 AUC: statistically significant on 443k loans but economically immaterial, while adding complexity and model-risk. Statistical significance is not the same as practical significance.

Q15. How did you tune hyper-parameters without overfitting to the test year?

Answer. Random search fitted on 2007–2014 with early stopping on 2015 (time-based). 2016 was used only to choose the champion and the recalibration; 2017 was untouched.

Q16. How would you explain a LightGBM PD to a regulator?

Answer. Global gain importance, SHAP values for individual decisions, monotone constraints if required, and a challenger WOE scorecard kept as the interpretable benchmark.

D. Calibration and backtesting

Q17. Discrimination is good; why do you still need calibration?

Answer. AUC is rank-based. ECL, capital and pricing use the PD level. Raw PDs averaged 4.8% against an observed 5.8% in 2017, i.e. ECL would be understated by about 20%.

Q18. Platt vs isotonic vs intercept shift: how did you choose?

Answer. By a rule fixed in advance on 2016: if the calibration slope differs from 1 (it was 0.92, $p \approx 10^{-26}$), use Platt; then an LR test for residual grade effects ($p = 10^{-19}$) justified Platt + grade.

Q19. Hosmer–Lemeshow rejects but you call the model calibrated. Why?

Answer. With 443k loans HL detects basis-point deviations. I report effect sizes: mean PD 5.90% vs 5.80%, ECE 0.24 pp, CITL -0.02 (conservative), Spiegelhalter $p = 0.08$, and no red grade in the Jeffreys test.

Q20. Explain the Jeffreys test.

Answer. With a Jeffreys prior $\text{Beta}(1/2, 1/2)$, the posterior of the default rate is $\text{Beta}(D + \frac{1}{2}, N - D + \frac{1}{2})$. The p -value is the posterior probability that the true DR is below the PD; small values mean the PD is too low. It is the ECB's recommended PD backtest.

Q21. What are PSI and CSI, and what thresholds apply?

Answer. $\sum (a - e) \ln(a/e)$ over bins: < 0.1 stable, 0.1 – 0.25 monitor, > 0.25 significant shift. Score PSI was 0.009 in 2017. CSI showed large values for fields missing before 2012, which I showed was a missing-bin artefact by switching to a 2013–2015 reference.

Q22. What is a vintage analysis and what did it show?

Answer. Cumulative default rate by months on book for each origination cohort. Vintages deteriorated from 2013 to 2016 (24-month CDR 9.2% $\rightarrow$ 12.3%), consistent with the higher default rates in the OOT years.

E. LGD, EAD and ECL**Q23. How did you compute LGD and why use the mean?**

Answer. $\text{LGD} = 1 - \text{net recoveries}/\text{EAD}$ on mature charge-offs (≥ 18 months). A fractional logit, segment means and LightGBM did not beat the pooled mean out-of-time (Diebold–Mariano). Recoveries come from bulk debt sales, which are not borrower-driven.

Q24. What is downturn LGD and how did you estimate it?

Answer. The LGD expected in an economic downturn (Basel requirement). I used the worst annual mean LGD among default years with at least 1,000 defaults: 0.936 (2012).

Q25. Why fractional logit for LGD and EAD?

Answer. The target lies in $[0, 1]$. Fractional logit keeps predictions in range and is consistent under quasi-likelihood with robust SEs (Papke–Wooldridge); OLS could predict outside $[0, 1]$.

Q26. What is the difference between EL and ECL?

Answer. Both are $\text{PD} \times \text{LGD} \times \text{EAD}$. Basel EL uses a through-the-cycle PD and downturn LGD over one year. IFRS 9 ECL is point-in-time, forward-looking, probability-weighted, and 12-month (stage 1) or lifetime (stages 2 and 3).

Q27. How did you obtain lifetime PD?

Answer. From matured vintages I estimated, per grade and term, $k = \ln(1 - \text{CDR}_{life})/\ln(1 - \text{CDR}_{12})$, a cumulative-hazard ratio, and set $\text{PD}_{life} = 1 - (1 - \text{PD}_{12})^k$.

Q28. How did you backtest the ECL?

Answer. Realised 12-month loss \$326.1M vs predicted \$321.5M (ratio 1.014). Placed in the correlated loss distribution, the realised loss sits at the 59th percentile ($p = 0.82$). An independence-only test would wrongly reject at 5%.

F. Portfolio model, VaR and ES**Q29. Explain the Vasicek one-factor model.**

Answer. Each obligor's asset value is a mix of a common factor and an idiosyncratic shock; defaults are independent given the factor. The conditional PD is $\Phi((\Phi^{-1}(\text{PD}) + \sqrt{\rho}z)/\sqrt{1 - \rho})$. It is the basis of the Basel IRB formula (ASRF).

Q30. How did you estimate asset correlation and why not use it?

Answer. MLE on 36 quarterly cohort default rates with the Vasicek density and a free level parameter, with a block-bootstrap CI: $\rho = 0.0037$. The window 2009–2017 has no recession, so it understates tail dependence. I used the Basel retail correlation (mean 0.065) for the base case and the data estimate as a sensitivity.

Q31. VaR vs Expected Shortfall?

Answer. VaR is a quantile and ignores losses beyond it; it is not sub-additive in general. ES is the average loss beyond VaR, is coherent, and is preferred by Basel FRTB. Base: 99% VaR \$723.7M, 99% ES \$820.6M.

Q32. How do you know the Monte Carlo is right?

Answer. Simulated EL matches the analytic EL ($p = 0.44$); simulated VaR is within 1% of the ASRF formula; VaR converged by 5k scenarios; VaR CIs from order statistics; unit tests on simulated portfolios.

Q33. Why did you simulate the whole portfolio and not 10k loans?

Answer. A sample changes the concentration (granularity) of the portfolio and the result does not scale to the real exposure; with only 2k scenarios the 99% VaR has about $\pm 8\%$ Monte Carlo error.

Q34. What is economic capital here?

Answer. Unexpected loss = $\text{VaR}_{99.9\%} - \text{EL} = \618.3M (9.4% of funded).

G. Stress testing**Q35. Why stress PD odds instead of multiplying PD?**

Answer. Multiplying PD can push it above 1 and stresses high-risk loans unrealistically; an odds multiplier keeps PDs in $(0, 1)$ and is equivalent to a shift of the logit.

Q36. What drives tail risk most in your results?

Answer. PD level and correlation. PD odds $\times 2$: +62% VaR; $\rho \times 1.5$: +18% VaR at the same EL; LGD stresses only about +5% because LGD is already about 0.9.

Q37. What is a reverse stress test?

Answer. Start from an outcome (loss exhausting capital) and solve for the scenario that causes it. Here PD odds must rise by $\times 3.74$ before expected loss equals the base 99.9% VaR.

Q38. How would you build a macro stress test with more data?

Answer. Regress the implied systematic factor $\hat{z}_t$ (or default rates) on macro variables (unemployment, GDP, rates); project them under regulatory scenarios (CCAR/EBA); map to conditional PDs and recompute losses.

H. General and behavioural**Q39. What was the most important finding of the project?**

Answer. That the original design was biased. Correcting the target changed the portfolio size from \$2.42bn to \$6.58bn and the ECL from a censored \$497M to a backtested \$321.5M (12-month).

Q40. What would you improve next?

Answer. Macro-economic drivers for point-in-time PD; SHAP-based explainability and monotone constraints; a survival model for lifetime PD; a PD–LGD correlation in the portfolio model; and a monitoring pack (PSI, backtests, champion–challenger).

Q41. Statistical vs practical significance – give an example from the project.

Answer. Stacking improved AUC by 0.0009 with $p < 10^{-6}$, and Hosmer–Lemeshow rejected with an ECE of only 0.24 pp. With 443k loans almost everything is significant; decisions must also consider effect size.